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Demand Forecasting & Inventory Management with Machine Learning in Retail Chains

The case of AutoLube Misr — an AI + ERP transformation of long-lead-time distribution in Egypt's automotive aftermarket

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24,000

active SKUs

330k+

part × location series

60–90 d

import lead time

>20%

interest rate

Structure of the case study



Section 3.1–3.4

Industry & business context



Section 3.5–3.8

Problem statement



Section 3.9–3.12

Data analytics component



Section 4

AI solution overview



Section 5

Analytical techniques applied



Section 6

Digital transformation analysis



Section 7

ERP integration & role



Section 8

Business impact & value creation



Section 9–10

Risks, ISO 27001 & recommendations

A parc-driven, inventory-intensive aftermarket — and a mid-sized B2B distributor inside it



Industry: automotive aftermarket (Egypt)

- Demand driven by the vehicle parc, not new-car sales
- >6 million vehicles; average age >10 years
- Heat, road conditions and import restrictions keep older cars running — and consuming parts
- Among the most inventory-intensive distribution sectors: long catalogues, sparse part-level demand
- Seasonality: summer heat, Ramadan & feast travel peaks

Source: McKinsey & Company (2018)

Organisation type: authorised distributor + multi-brand wholesaler (B2B)

24,000

active SKUs

2 DCs + 12

branches (14 locations)

4,800

B2B customers (workshops, retailers, fleets)

850

employees

EGP 3.6 bn

annual revenue

Oracle Fusion

Cloud ERP & SCM backbone

Composite firm (Yin, 2018) · data simulated for academic purposes

Five pre-adoption challenges produced a dual failure: too much of the wrong stock, too little of the right



Manual forecasting

10 planners · monthly Excel
· brand/category level — far
above part × branch



Lost sales

82% line fill rate —
workshops buy the missing
part from a competitor



Overstock

128 days of inventory; over-
ordering against a 60–90 d
lead time at >20% interest



Obsolescence

4.1% of inventory value
written off yearly
(superseded & slow parts)



Idle data & seasonality

Heat & Ramadan peaks by
intuition; ERP data only for
backward reports

Dual failure of long-lead-time distribution overstock of the wrong items + stock-outs of the right ones, at the same time. Lost sales ≈ EGP 130M/yr and holding cost ≈ EGP 220M/yr (simulated baseline).

Strong structural demand, hostile operating conditions, and a squeeze from informal competition



Structural demand

- Parc >6M vehicles, ageing
- Harsh climate → higher wear
- Import limits keep old cars in use
- Seasonal peaks (heat, Ramadan)



Operating frictions

- FX depreciation → landed-cost shocks
- Letters-of-credit disruption
- Customs → lead-time variance
- Inflation → cash-sensitive buyers



Competitive rivalry

- Other authorised distributors
- Parallel importers
- Informal channel & counterfeits
- Low switching cost for workshops

✗ Win by holding more stock?

No — >20% financing cost and obsolescence forbid it.

✗ Win by raising prices?

No — customers defect to parallel/informal channels.

✓ Source of advantage

Predicting part-level demand over the 60–90-day import lead time — and acting on it faster than rivals.

The business problem, clearly defined

“

AutoLube Misr could not predict demand at part-number × branch × week level over its 60–90-day import lead time — and so could not order the right products, in the right quantities, for the right locations, months in advance.

55%

forecast accuracy (1 – WMAPE)

82%

line fill rate

128 d

inventory days of supply

4.1%

annual obsolescence

Research questions derived from the problem

RQ1

Can ML forecasting, integrated with ERP, overcome judgemental forecasting under long lead times and intermittent demand?

RQ2

Which components — forecast, inventory policy, transfers, process automation — account for the gains?

RQ3

How robust are the gains under shocks typical of an emerging economy (FX, customs, supplier disruption)?

Four structural weaknesses defeated spreadsheet judgement — the ERP was not the weak link, its forecast input was



Scale

330,000+

SKU × location series — beyond manual planning capacity (~33k per planner at best)



Intermittency

Sparse

most parts sell in small, irregular quantities; moving averages fail (Syntetos et al., 2005)



Non-linearity

Interacting

drivers: parc mix × heat × holidays × fuel × FX-driven prices



Latency

6 d + 60–90 d

manual planning cycle on top of the import lead time



Diagnosis: Oracle's supply-planning logic was sound — the forecast feeding it was the weak link: wrong granularity (brand vs part × branch × week), wrong horizon (month vs full lead time), no modelled seasonality or drivers.

The gap typology: every failure has a measurable cost

Gap type	Description	Quantified impact (simulated baseline)
Operational	82% line fill; overstock of slow movers; 6-day planning cycle on top of 60–90-day lead times; planners absorbed by data entry	18% of order lines unfilled · 128 days of inventory · ~70% of planner time on manual tasks
Strategic	Cannot guarantee the same-day availability workshops expect; exposed to parallel importers & counterfeits; no data basis for regional range decisions	~2% annual B2B customer churn linked to availability
Financial	Working capital locked in the wrong stock; obsolescence write-offs; lost sales; emergency air-freight expediting	Obsolescence 4.1% of inventory value · lost sales ≈ EGP 130M/yr · holding cost ≈ EGP 220M/yr · air freight 9% of inbound

Reading: the three gap types share one root cause — a forecast at the wrong grain and horizon — which is why a single AI + ERP intervention can close all three.

Four data types, six source systems, one feature store

Data type	Examples	Source system	Latency
Structured	Sales orders, invoices, stock positions, import POs, part master with vehicle fitment, supplier lead times	Oracle Fusion ERP/SCM (OM, Procurement, Inventory, Financials)	Nightly batch
Unstructured	WhatsApp/email orders, complaint tickets, mechanic-forum & social-media posts, supplier correspondence	CRM, messaging, web — processed with NLP	Daily
Real-time	Branch sell-in, key-account sell-out, warehouse scanner events, container tracking	Sell-out feeds, Oracle WMS, logistics providers (IoT)	Streaming
Historical & external	5 years of sales, prices, FX, promotions, weather, holidays, vehicle registrations	Data lake; public statistics; weather services	History + daily

Brief's source categories — coverage

**ERP systems**
Oracle Fusion Cloud

**CRM**
4,800 accounts; segments, credit

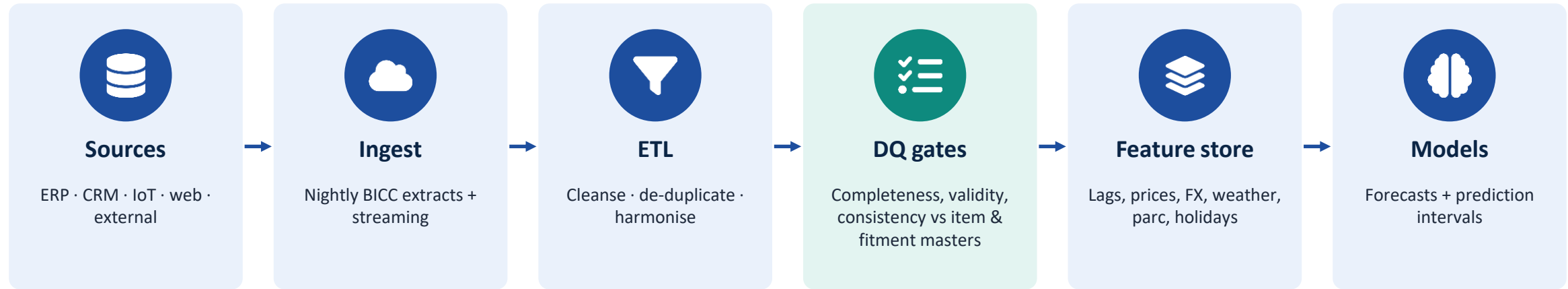
**IoT**
Scanners, containers, telematics pilot

**Social media**
Mechanic forums, posts (NLP)

**Transaction data**
Orders, invoices, receipts

Leading indicators with most signal: vehicle-registration mix (parc) and temperature forecasts.

Pipeline and data quality: forecast quality is bounded by data quality



3 months

master-data cleansing before go-live: missing lead times, broken supersession chains, duplicates, incomplete fitment

76.1%

forecast accuracy under poor data quality (S13) vs 82.9% in S5 — a bigger hit than any single external shock

Nightly

actuals flow back to retrain; weekly model retraining; MLOps drift monitoring closes the loop

Machine learning at the core: segment demand first, then match the method to the pattern

Demand class	Value share	Primary method	Rationale
Fast-moving lubricant	44.5%	LSTM + GBM	Rich seasonality; exogenous drivers
Seasonal lubricant	13.1%	LSTM	Heat & Ramadan cycles
Regular spare part	30.8%	Gradient boosting (XGBoost / LightGBM)	Price × FX × parc interactions
Superseded / new	2.4%	k-means fitment inheritance	No or broken own history
Intermittent · lumpy · slow	9.0%	Croston / TSB + safety stock	Zero-inflated; bias-corrected

Forecast target: cumulative demand over each supplier's lead time, with prediction intervals → statistical, part-level safety stock instead of uniform padding. Model selection per SKU by rolling-origin back-testing.



Machine learning

GBM · LSTM · Croston/TSB · k-means



NLP

Intent, complaint & counterfeit signals from text



Computer vision (pilot)

Reads part labels at goods receipt



Optimisation

Safety stock, order quantities, transfers

Five business functions change how they decide



Primary impact: supply chain (dark). Secondary impacts flow through shared ERP data — the same forecast drives sales promises, finance plans and warehouse schedules.

Each structural weakness is matched by a specific AI mechanism — and a measurable result

Weakness	AI mechanism	Result
Scale	Trained model scores 330k part-branch series in minutes; per-SKU model selection	Planner coverage ~33k → ~110k series
Intermittency	Croston/TSB for sparse tail; k-means fitment clustering pools similar parts	Superseded 35% → 65%; new parts 2% → 34% accuracy (sim.)
Non-linearity	Tree ensembles learn interactions (heat × batteries; price-rise pull-forward)	Accuracy 55% → 82% (narrative)
Latency	Cumulative lead-time forecast + prediction intervals; nightly push to Oracle; RPA release	Replenishment cycle 6 d → <1 d

Key design choice: forecasting and inventory theory are coupled — the prediction interval, not the point forecast, sets the buffer.

Supervised, unsupervised and semi-supervised learning — each does a distinct job in the pipeline



Supervised

Learns a mapping from labelled inputs to a known target.

ALGORITHM

Regression: XGBoost / LightGBM, LSTM, Croston/TSB

USE IN THIS CASE

Forecast units per part × branch × week from lags, price, FX, weather, holidays

OUTPUT

Point forecast + prediction interval



Unsupervised

Finds structure in unlabelled data.

ALGORITHM

k-means clustering (ADI / CV² features, fitment attributes)

USE IN THIS CASE

Segment SKUs by demand pattern; cluster branches by parc; new parts inherit forecasts

OUTPUT

Segments → model routing



Semi-supervised

Few labels + many unlabelled examples.

ALGORITHM

Label propagation on text embeddings

USE IN THIS CASE

Few thousand labelled messages → millions: intent, complaint, counterfeit

OUTPUT

Early demand & risk signals

Chained pipeline: unsupervised segmentation → routes each series to a supervised model ← semi-supervised text signals as features

RPA is the ‘hands’ that turn AI decisions into ERP transactions

Bot sequence (UiPath via Oracle Integration Cloud)

- 1 Validate nightly data loads
- 2 Convert approved forecasts → import PRs in Oracle Procurement (auto-release within tolerance)
- 3 Route exceptions to planners
- 4 Send order confirmations to suppliers
- 5 Update container milestones & ETAs
- 6 Reconcile goods receipts vs POs; daily exception report

	RPA	Machine learning
Logic	Deterministic rules	Probabilistic, learned
Learns?	No	Yes — retrained weekly
Role	‘Hands’: executes	‘Brain’: predicts

Together = intelligent process automation (Lacity & Willcocks, 2021)

~70%

planner manual
workload removed

7 → 2

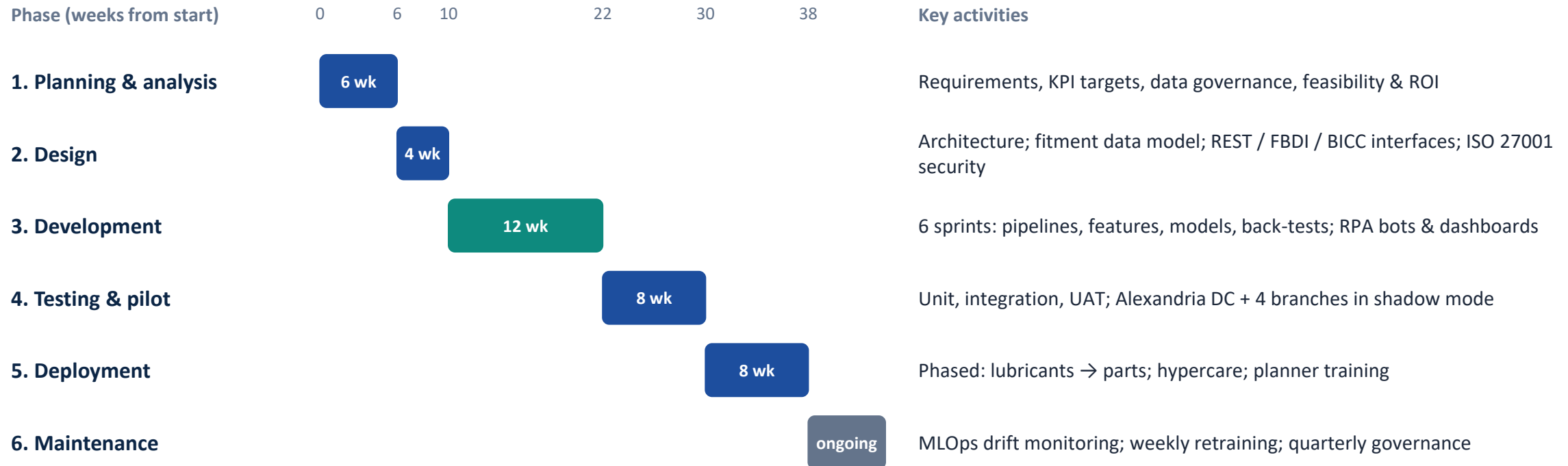
process hand-offs

6 d → <1 d

replenishment cycle

Controls: dedicated least-privilege bot accounts, value tolerances, segregation of duties, annual internal-audit review.

A hybrid SDLC: Agile for the models, stage-gates for the ERP, CRISP-DM for the data science



Agile Scrum · AI components

Requirements evolved as planners learned what models could do; 2-week sprints



Stage-gate · ERP integration

Strict change control: an error in automated purchasing costs money immediately



CRISP-DM · Data science

Business → data understanding → preparation → modelling → evaluation → deployment

All four levels of analytics — from reporting what happened to recommending what to do

Value & difficulty ↑

Before AI: descriptive only (monthly backward-looking reports). After: all four levels, nightly.



Descriptive

What happened?

Daily dashboards: sales, fill rate, stock, ageing & obsolescence risk by branch and part



Diagnostic

Why did it happen?

Drill-downs attributing variances to stock-outs, price changes, competitor promotions, late containers



Predictive

What will happen?

Part × branch × week forecasts with prediction intervals; promotion uplift; obsolescence risk



Prescriptive

What should we do?

Import quantities, dynamic safety stock, allocation, transfers, clearance pricing — executed by RPA

A balanced KPI scorecard — each metric guards against gaming another

KPI	Definition	Baseline	Year 1	Δ
Forecast accuracy	1 – WMAPE, part × branch × month	55%	82%	+27 pts
Line fill rate	% order lines shipped complete same/next day	82%	95%	+13 pts
Inventory days of supply	Average inventory ÷ daily COGS	128	86	–33%
Obsolescence write-offs	Annual write-offs, % of average inventory value	4.1%	1.9%	–54%
Emergency air freight	Expedited freight, % of inbound freight cost	9%	3%	–67%
Planner productivity	Part-branch combinations per planner	~33,000	~110,000	×3.3
Replenishment cycle	Data cut-off → import PO release	6 days	<1 day	–85%+

Leading

accuracy, forecast bias, data-quality
pass rate, override rate

Lagging

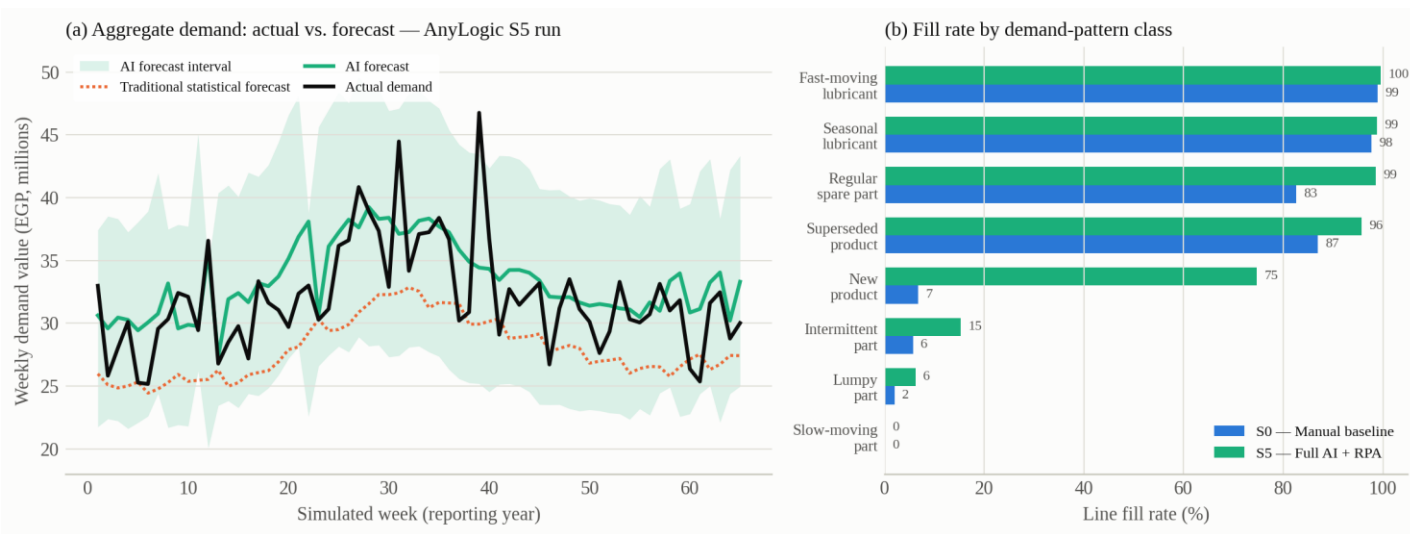
fill rate, inventory days, obsolescence,
ROI

Model health

drift (accuracy decay), retraining
cadence, interval coverage

Caveat: WMAPE degenerates at zero demand → MASE / service metrics for sparse parts

Sample dashboards: the planner's S&OP view and the operational simulation console



(a) weekly demand vs AI & statistical forecasts · (b) fill rate by class, S0 vs S5



AnyLogic operational console, S5, day 296 (single run)

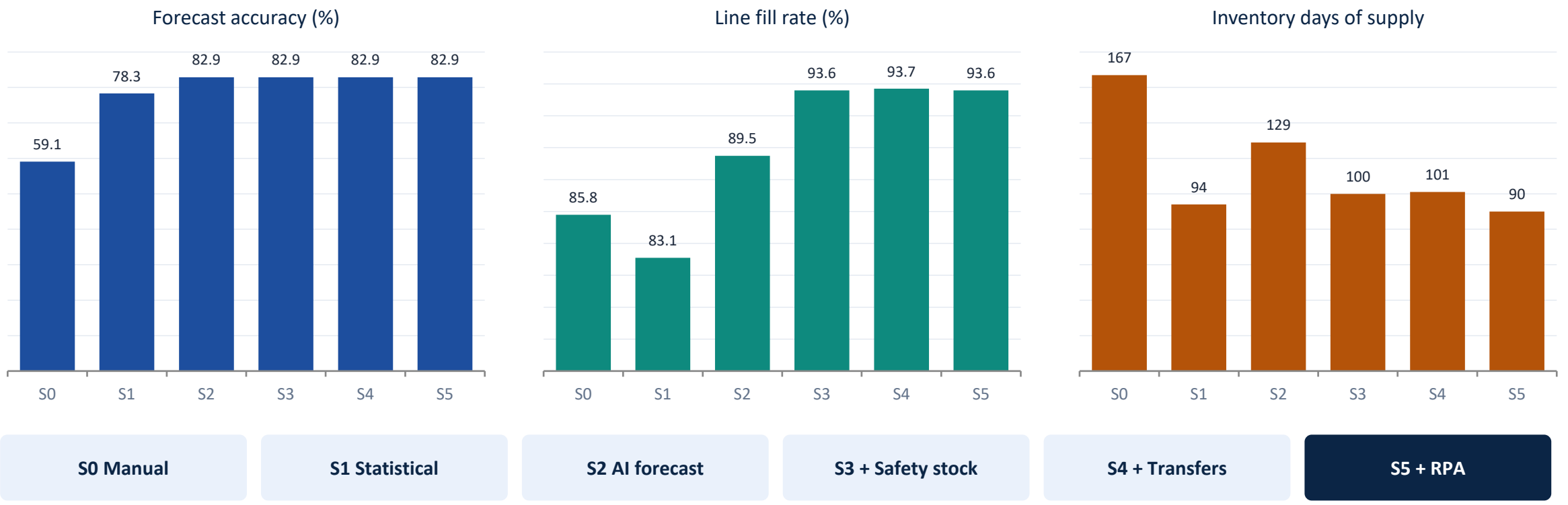
AI tracks peaks
the statistical forecast under-reacts to; the interval feeds safety stock

Service gains are uneven
regular 83→99%, superseded 87→96%, new 7→75%; sparse tail stays hard

Governance built in
drift checks and overrides logged on the same screen planners use



Adding one lever at a time: accuracy saturates early, cash arrives late



+19.2 pts S0→S1: discipline, not ML, drives most accuracy

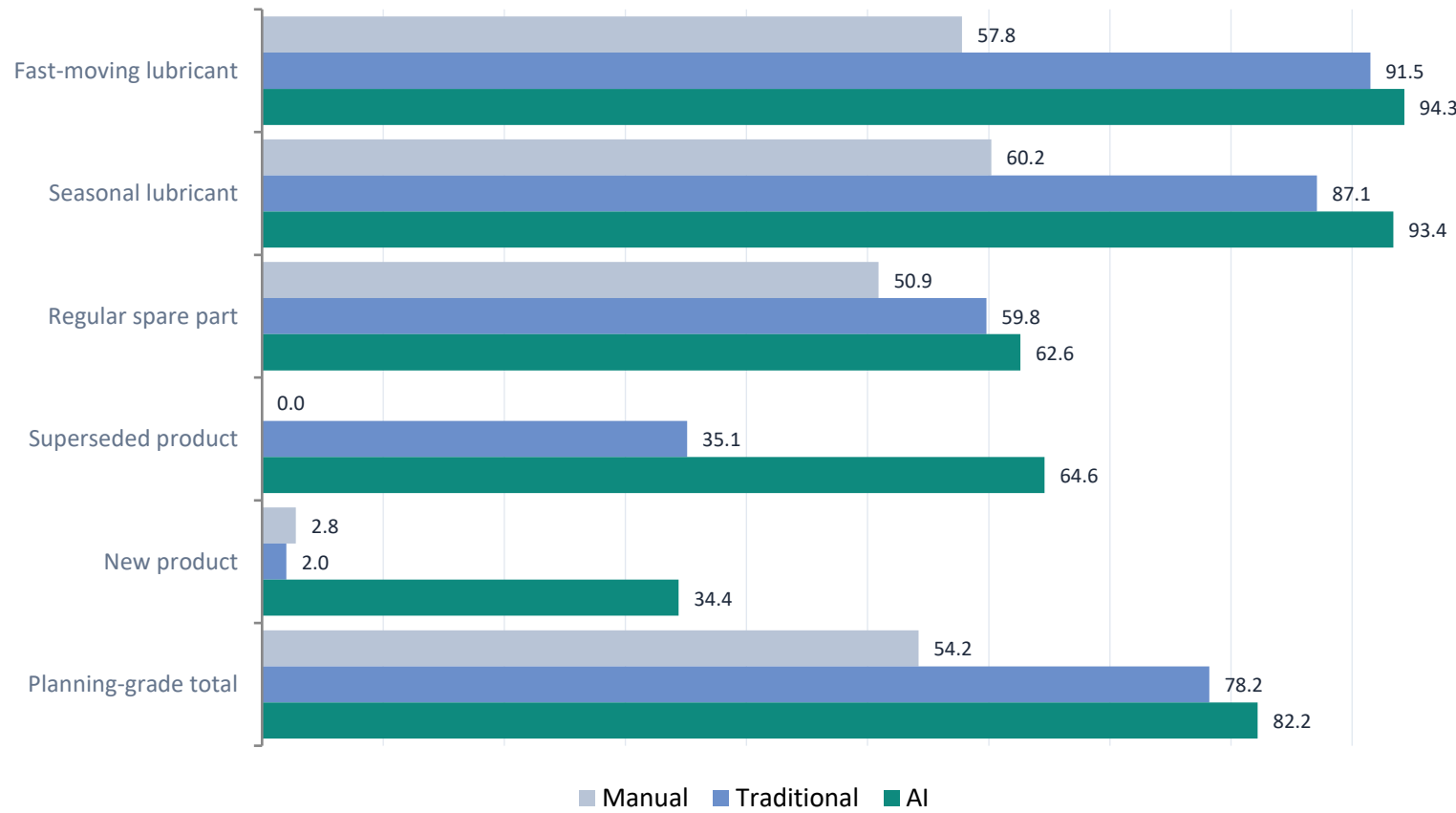
+4.6 pts S1→S2: the ML increment

57 → 130 net benefit EGP M, S2 → S5: complements > algorithm

Mean of 10 replications, paired CRN; 95% CI: accuracy ±0.3–1.6, fill ±0.5–0.6, inventory ±1–5 days.

ML's edge is concentrated where history is missing

Forecast accuracy (1 – WMAPE, %) by demand class



+29.5 / +32.4

pts AI over traditional for superseded / new products — via fitment-cluster inheritance

≈ 0%

accuracy for every method on intermittent, lumpy & slow classes (9% of value, most SKUs): the metric degenerates — manage by service level & safety stock

Data mining turns 24,000 items into four kinds of evidence-based decision



Classification Decision trees · logistic regression

Rank parts by stock-out / obsolescence risk → planners work a short exception list



Clustering k-means

Segment SKUs, branches, customers → model routing & par-based assortments



Association rules Apriori (support · confidence · lift)

Oil + oil/air filters; pads + discs → joint ordering, service kits, cross-sell



Regression Elasticity models

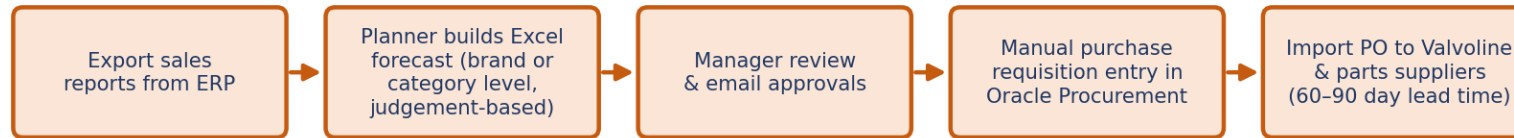
Price, FX & promotion elasticity → pricing, promotion and clearance decisions



Intuition-first → evidence-first decisions (Provost & Fawcett, 2013)

Import replenishment redesigned: seven hand-offs become two, six days become one night

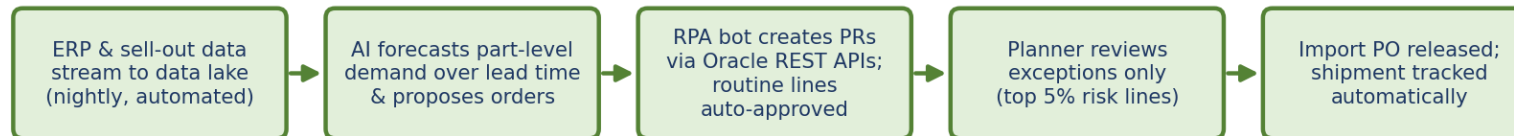
BEFORE (manual, ~6 days per cycle)



7 → 2
hand-offs

6 d → <1 d
cycle time

AFTER (AI-driven, same-day cycle)



Monthly → nightly
brand → part × branch × week

Email chains → tolerance
exception-based approval

Event-driven
heat wave or ship delay → same-day replan

Key process changes: (1) hand-offs reduced from 7 to 2; (2) approval loop replaced by exception-based control; (3) forecast granularity moved from brand/month to part-number/branch/week over the full import lead time; (4) cycle time cut from ~6 days to <1 day; (5) planners redeployed to supplier collaboration and range decisions.

Original process model (report Fig. 2)

Automation freed the planners; the harder change was cultural

Automation of manual, repetitive tasks

-  Report exports & spreadsheet consolidation
-  Purchase-requisition keying
-  Supplier confirmations & container chasing
-  Goods-receipt / PO reconciliation
-  Daily exception reporting

~70%

planner hours released

0

headcount cut — redeployed

Culture & decision-making

Override paradox: planners distrusted the 'black box'; most early overrides worsened results.



Explainability — each forecast shows its main drivers



Override-log review — monthly, reasons recorded



Retraining — planners → demand analysts

New norms: meetings start from the dashboard · dissent must be evidence-backed · management by exception.

For workshops and fleets, transformation shows up as one thing: reliability

82% → 95%

line fill rate

~2% → ~1%

availability-related B2B churn (halved)

Same day

availability for most lines



Reliable promises

Sales staff quote availability dates from live stock and inbound data



Local relevance

Branch assortments tuned to the local vehicle parc



Proactive service

Key accounts warned before shortages; alternatives offered



Trust & authenticity

Authorised supply vs counterfeits; CV pilot flags suspect labels

Before vs after: from spreadsheet judgement to evidence-based, automated planning

Dimension	Before (manual)	After (AI + ERP + RPA)
Forecasting	Excel · monthly · brand/category level · judgement	ML · nightly · part × branch × week over lead time · evidence
Replenishment	Manual PR entry · 6-day cycle	RPA-automated · same day · exception-based
Shipment visibility	Phone calls & emailed spreadsheets	Automated container tracking into Oracle & dashboards
Data use	Backward-looking monthly reports	Real-time predictive & prescriptive dashboards
Planner role	Data entry & firefighting	Analysis, supplier collaboration, range decisions
Customer experience	82% fill · unreliable promises	95% fill · proactive availability management

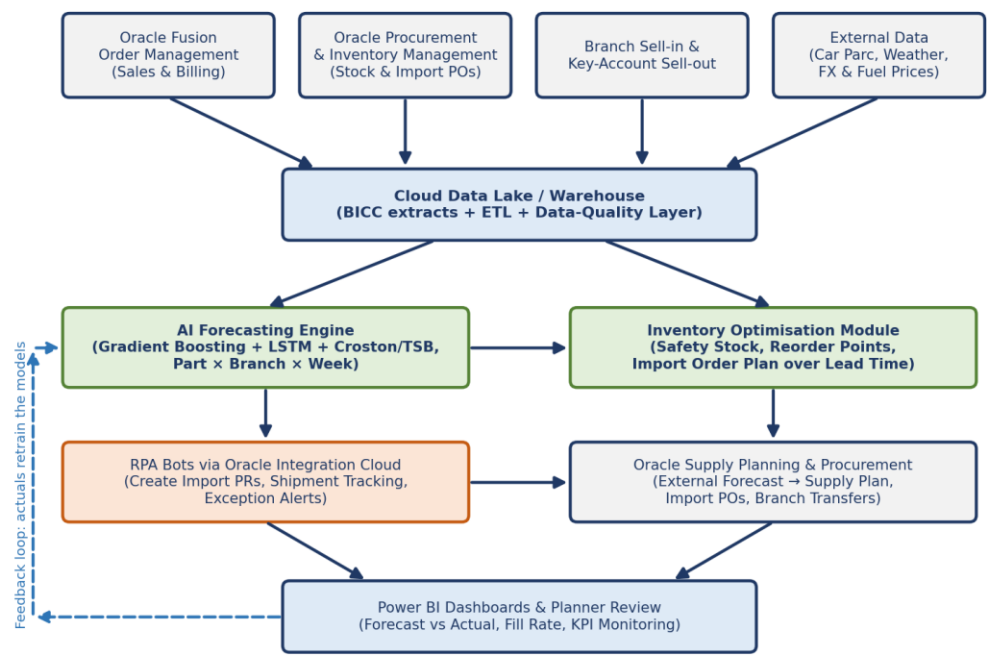
Transformation = redefinition of work, decision rights and value proposition, not only technology (Vial, 2019).

Five Oracle Fusion Cloud modules form the transactional backbone the AI plugs into



Oracle Fusion is SaaS: quarterly updates, core cannot be modified → intelligence lives outside the ERP and integrates through standard interfaces.

Hub-and-spoke: ERP as system of record, AI as an intelligent satellite

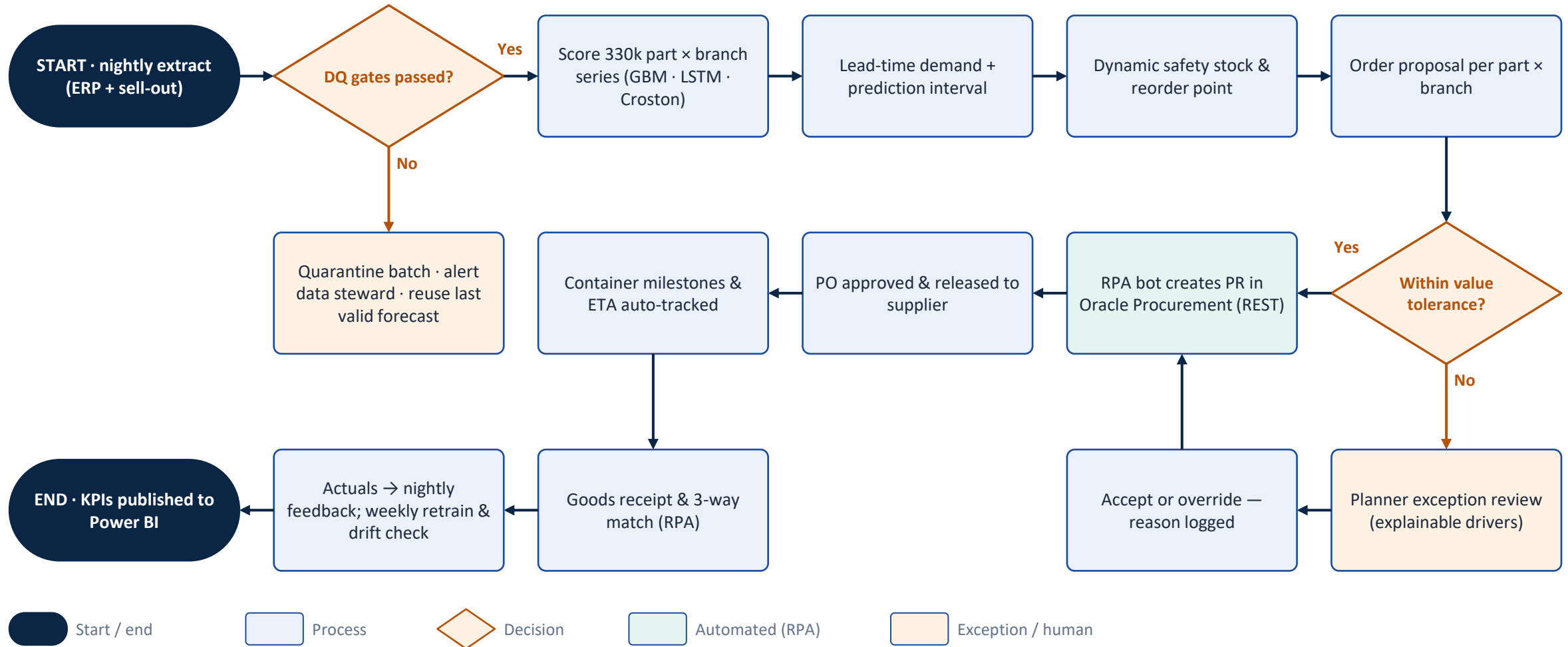


Original architecture diagram (report Fig. 3)

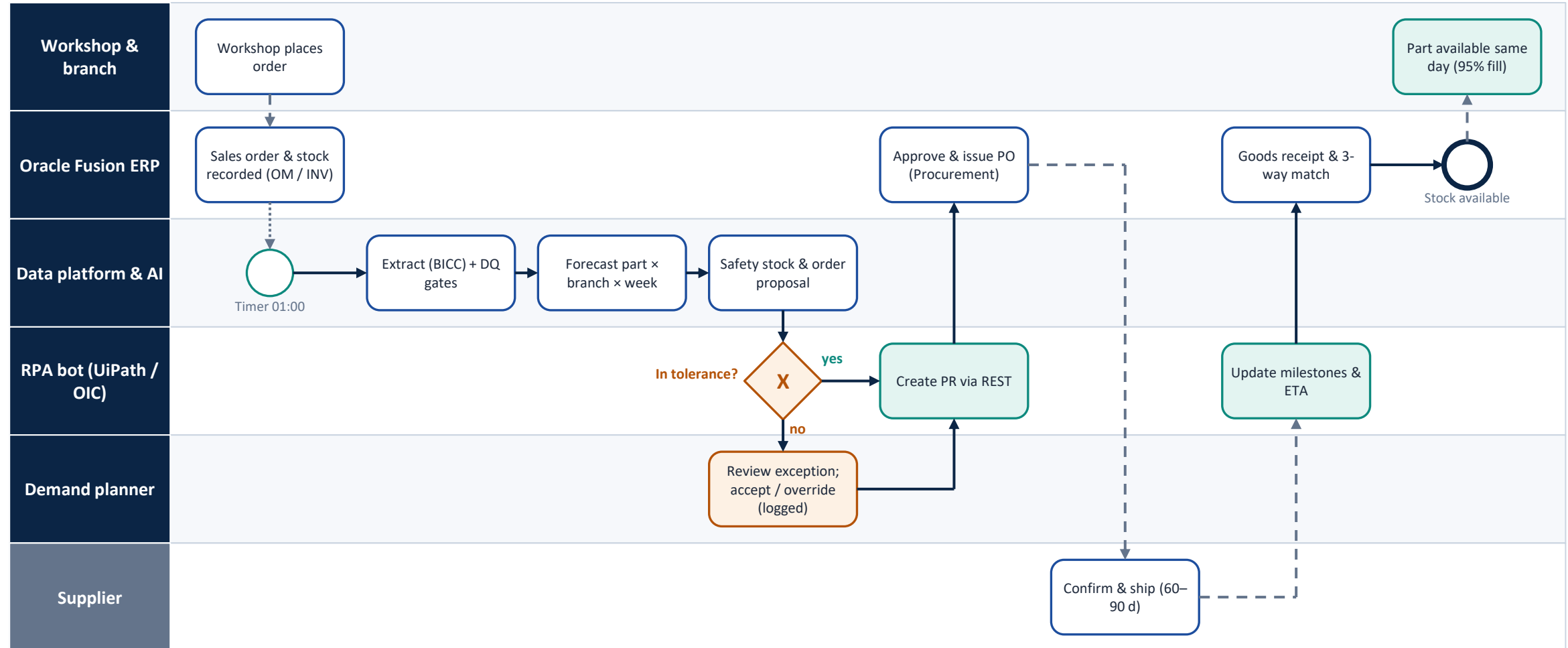
Flow	Mechanism	Frequency
ERP → data lake	Oracle BI Cloud Connector (BICC) extracts	Nightly
Sell-out, WMS, logistics → lake	Streaming interfaces / APIs	Near real-time
AI → Demand Management	External forecast load (FBDI)	Nightly
AI → Procurement	RPA creates PRs via REST APIs (OIC + UiPath)	Nightly / same day
ERP → dashboards	Power BI on lake + ERP KPIs	Daily
Actuals → models	Feedback loop; retrain & drift check	Nightly / weekly

Design principles: SaaS core untouched · configuration + REST + FBDI + OIC only · one system of record · full audit trail (POs, receipts, invoices stay in standard ERP flow).

Flowchart: the nightly AI-ERP replenishment decision cycle

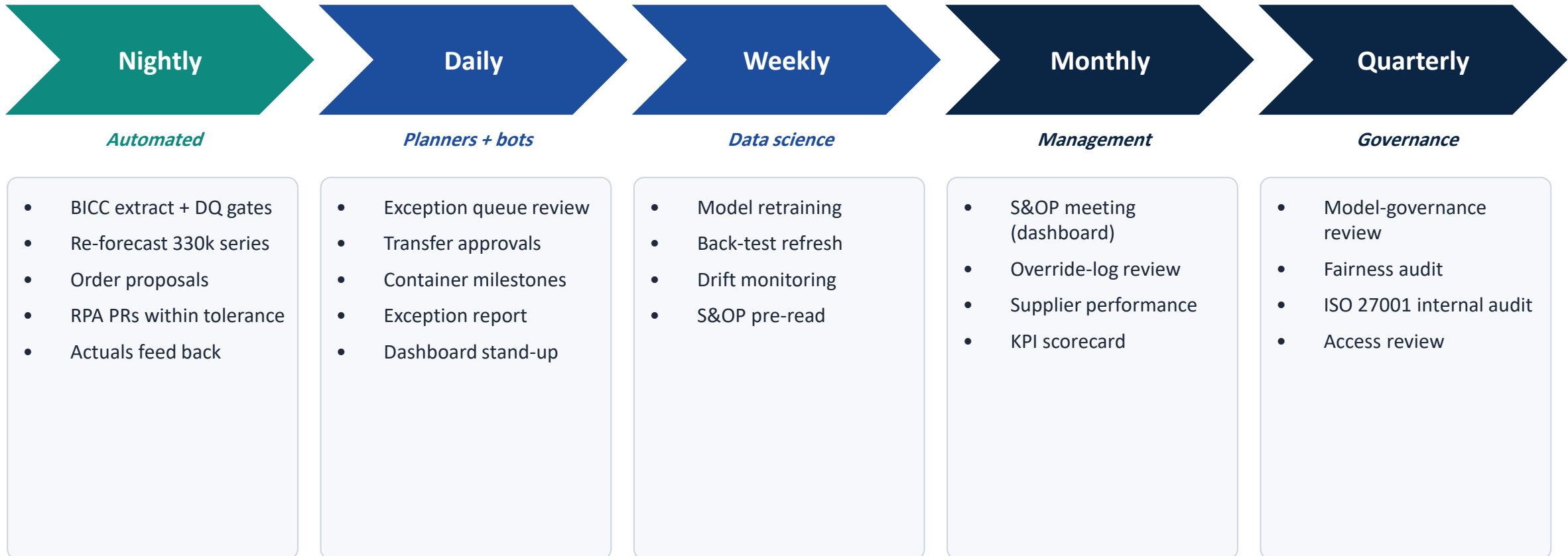


BPMN: import replenishment 'to-be' process across six participants



○ timer start ● end event ◇ X exclusive gateway → sequence flow - - → message flow ····→ data association

Workflows: who does what, how often — from nightly automation to quarterly governance



Automation handles volume; people handle exceptions and governance.

Integration creates a reinforcing loop — but it was also the riskiest workstream



Benefits of ERP + AI

Single source of truth

One system of record; AI never bypasses ERP

Straight-through processing

No re-keying across 24,000 part numbers

Auditability

Import & customs documents fully traceable

Reinforcing loop

Master-data discipline ↑ accuracy; anomaly detection exposes data errors



Implementation challenges

Master-data quality

Missing lead times, broken supersession, duplicates → 3 months cleansing

Batch windows

Nightly extracts tuned to fit the window

SaaS rigidity

Quarterly updates; no core customisation → REST / FBDI / OIC only

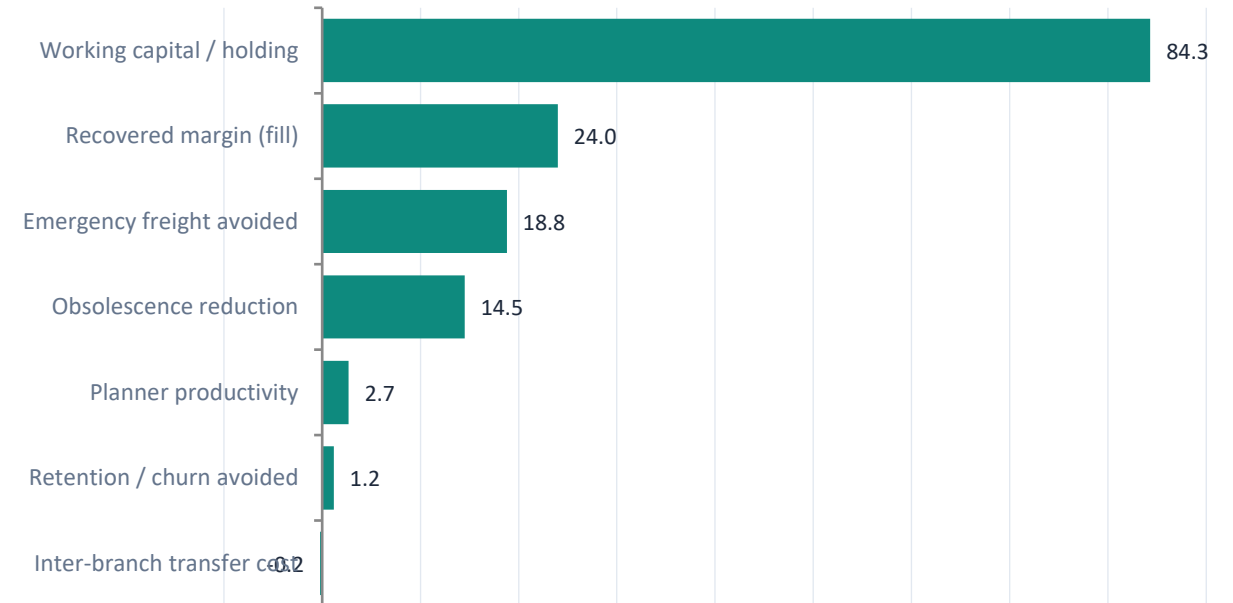
Skills & resistance

Scarce Oracle integration skills; user resistance

Financial impact: a working-capital problem disguised as a forecasting problem

Value driver (narrative)	EGP M / yr
Working-capital release (128→86 d; ~EGP 340M at ~22%)	74
Obsolescence reduction (4.1% → 1.9%)	22
Lost-sales recovery (fill 82% → 95%)	18
Expediting savings (air freight 9% → 3%)	6
Planner productivity (70% automated)	5
Total annual benefit	≈ 125
Year-1 cost (build + run)	50
Year-1 ROI · payback	≈150% · ~5 mo

Simulation: annual benefit S5 vs S0 (EGP M, mean of 10 reps)



58%
of gross benefit from working
capital

145 → 130
gross → net EGP M

≈191%
Year-1 ROI · payback 3.2 mo

Operational efficiency: faster cycles, leaner stock, more coverage per planner



6 d → <1 d

replenishment cycle



33k → 110k

series per planner (×3.3)



128 → 86 d

inventory days of supply



4.1% → 1.9%

obsolescence write-offs



9% → 3%

emergency air freight share



Optimiser

proposes inter-branch transfers

Narrative Year-1 estimates; simulation agrees on direction and magnitude (e.g., inventory 167 → 90 days, -46%).

Strategic advantage lives in the bundle, not the model: sense, seize, reconfigure



Sense

Nightly re-forecasting · NLP demand signals · drift monitoring

Evidence: S2, S12



Seize

RPA auto-release · same-day open-order adjustment

Evidence: S5, shock tests



Reconfigure

Dynamic safety stock · inter-branch transfers · parc-based ranges

Evidence: S3, S4

Dynamic capabilities (Teece, 2007)

Imitable

The ML algorithms themselves — commodity tools (GBM, LSTM).

Hard to imitate

Cleansed master data + integrated processes + analytically literate planners.

< 50%

of net value from the algorithm alone (S2 EGP 57M vs S5 EGP 130M)

Scalable, sustainable — and most valuable precisely when conditions are worst



Scales at marginal cost

New branches, SKUs, brands on the same cloud platform



Recurring benefit

EGP 15M/yr run cost vs EGP 130M net benefit (sim.)



Environmental

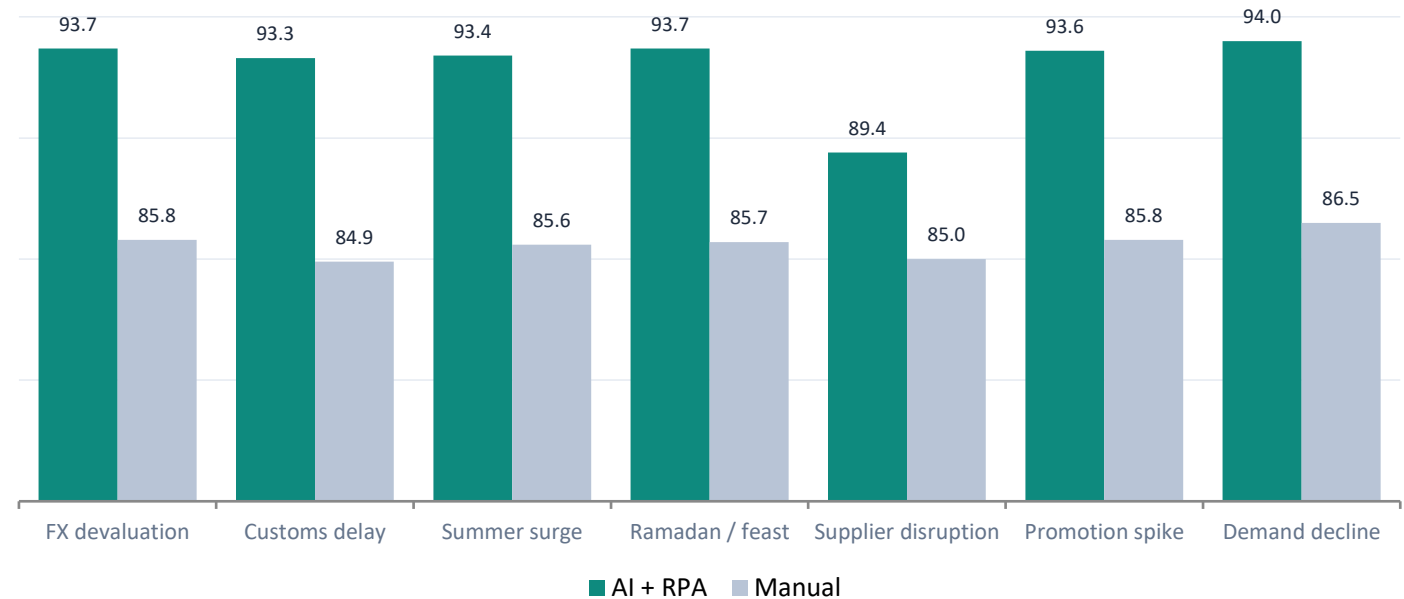
Less obsolescence waste and air-freight emissions



Resilient

Advantage holds under every single modelled shock

Line fill rate under shock (%) — same shock, 10 reps each



Limit: when shocks coincide (\$15) net benefit falls to ≈ EGP 3M — robust to single shocks, fragile to compound ones.

Privacy: B2B forecasting still processes personal data — so PDPL and GDPR apply

Personal data in scope



Workshop owners' contact & behavioural records



Credit information (CRM)



Telematics pilot: vehicle usage linked to drivers



Customer messages processed with NLP

Legal basis: Egypt PDPL Law No. 151/2020; GDPR (EU 2016/679) where EU-linked partners are involved.

Principles and controls applied

Lawful basis & consent

documented per processing purpose

Purpose limitation

forecasting ≠ marketing without new basis

Data minimisation

pseudonymised features in the feature store

Storage limitation

retention schedules per data class

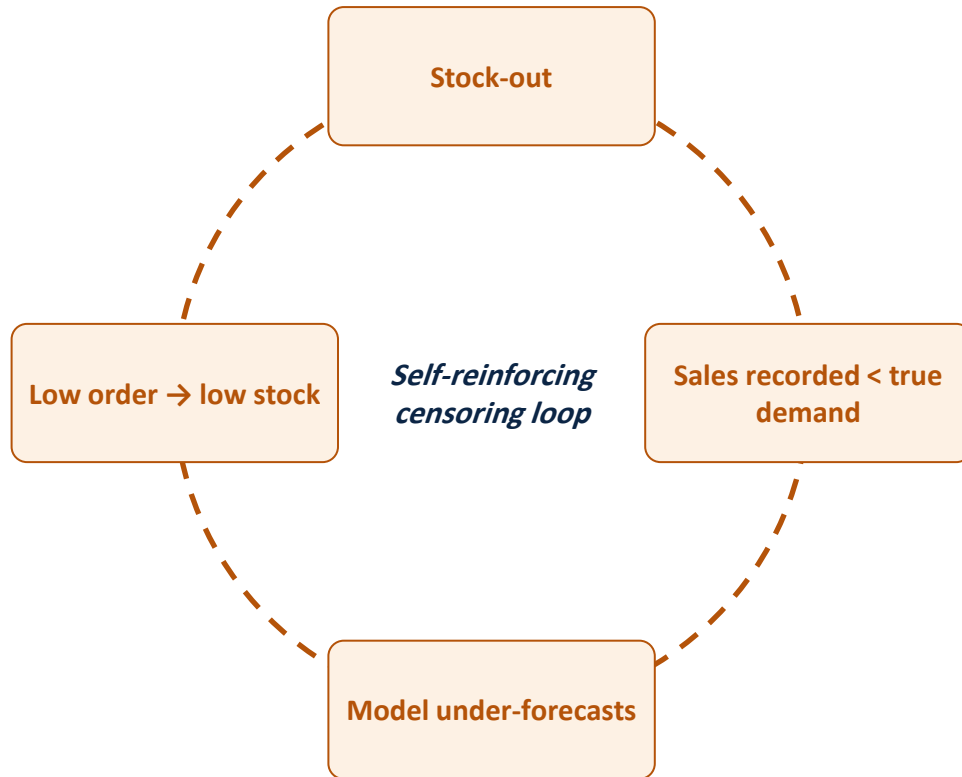
DPIAs

before go-live and for the telematics pilot

Security

encryption · role-based access · audit logging

Bias: censored demand, allocation fairness and human override bias



Fix: reconstruct true demand from unfilled order lines



Allocation fairness

Risk: Revenue-following allocation starves rural branches & small workshops

Mitigation: Branch fill-rate floors; audits of customer-scoring models



Human bias in the loop

Risk: Algorithm aversion → frequent overrides, most of which worsened results

Mitigation: Explainability, override-log review; watch for automation bias too



Data & label bias

Risk: Hand-labelled text seeds label propagation — errors propagate at scale

Mitigation: Stratified labelling, periodic re-labelling and audit samples

Compliance obligations and adoption risks — quantified where the simulation allows

Regulatory & compliance

 Customs documentation & valuation

 Egypt e-invoicing regime

 Warranty & consumer protection

 Anti-counterfeit enforcement

 Financial controls: bot SoD, tolerances, annual audit

Adoption risk	Evidence	Level	Mitigation
Change resistance	Most early overrides worsened results	Med	Explainability, training, override board
Model drift	S12: 82.9% → 80.5%; ≈ -EGP 10M	Med	Weekly retraining; MLOps drift alerts
Data quality	S13: accuracy 76.1%; net ≈ EGP 93M	High	DQ gates; data-governance office
Integration fragility	Bad load / lead time → large orders	High	DQ gates; value tolerances; kill switch
Skills dependency	Vendors; scarce data-science talent	Med	Internal analytics academy
Compound shocks	S15: net ≈ EGP 3M; ROI ≈ -64%	High	Scenario planning, hedging, digital twin

Limitations: what the system cannot do — and what this study cannot claim



Limitations of AI & ERP systems

No foresight of the unprecedented

import bans, sudden devaluation → hedging & scenario planning still required

Sparse tail stays hard

point accuracy ≈ 0 for intermittent parts; safety stock carries service

Probabilistic output

planners must keep judgement to challenge the model

ERP rigidity

SaaS core, quarterly updates, nightly batch windows limit real-time response

Master-data cost

data discipline is a permanent operating cost



Threats to validity (this study)

External validity

composite firm, synthetic data → plausible magnitudes, not estimates; ROI conditional on >20% rates

Construct validity

1 – WMAPE unstable at zero demand → MASE / cost-service metrics

Internal validity

paired CRN + 10 reps support attribution; fixed lever order $\neq$ Shapley

Independence validity

case & simulation share assumptions → corroboration, not replication

ISO/IEC 27001:2022: the clauses and controls most relevant to an AI-ERP platform

ISO/IEC 27001:2022 reference	Requirement	Application in the AutoLube Misr AI-ERP platform
Clause 6.1.2 / 8.2	Risk assessment & treatment	DPIA and security risk assessment of data lake & AI pipeline before go-live
A.5.9 / A.5.12	Asset inventory & classification	Pricing, cost, customer & telematics data classified; CRM data marked personal/confidential
A.5.15 / A.8.2	Access control & privileged access	Role-based access to the data lake; RPA bots on least-privilege accounts with value tolerances
A.8.24	Use of cryptography	Encryption at rest and in transit (Oracle ↔ data platform ↔ supplier interfaces)
A.5.19 – A.5.23	Supplier & cloud-service security	Security clauses in cloud and AI vendor contracts; risk assessment of tracking & telematics providers
A.8.16	Monitoring activities	Logging & alerting on pipelines, model APIs and bot purchasing transactions
A.5.24 – A.5.26	Incident management	Response plan for data breaches or malfunctioning automated ordering

AI-specific point: a malfunctioning ordering bot is treated as a security incident, not only an IT fault.

Five strategic recommendations, each targeting a binding constraint the evidence identified

01



Data-governance office

Own part masters, supersession chains & fitment catalogue

ADDRESSES
Binding constraint: S13 data quality

02



Deeper demand sensing

Expand sell-out & POS capture from key accounts

ADDRESSES
Earlier signals → less buffer

03



CPFR with suppliers

Share forecasts with Valvoline & major vendors

ADDRESSES
Shorter effective lead time; less bullwhip

04



Formal AI governance

Model-risk policy, model cards, fairness audits, override board

ADDRESSES
Override & bias risks

05



Analytics academy

Internal capability that survives staff & vendor changes

ADDRESSES
Skills-dependency risk

An AI upgrade roadmap across three horizons



Horizon timing is a proposed sequencing.

From an inventory project to an intelligent-enterprise operating model



Digital transformation opportunities

- B2B e-commerce portal with live availability
- Computer-vision counterfeit screening
- Delivery-route optimisation
- Credit-risk & loyalty analytics



Expansion of ERP-AI capabilities

- Evaluate Oracle Demand Mgmt built-in ML
- Oracle Fusion embedded AI agents
- Finance: cash-flow & FX-exposure forecasting
- HR: shift scheduling



Research agenda (PhD)

- Replicate on real multi-firm spare-parts panels
- Value-based (cost-service) forecast evaluation
- Shapley / full-factorial lever attribution
- Field experiments on explainability & overrides

CONCLUSION

When goods take three months to arrive, prediction quality is the business — but prediction alone is not the value.



RQ1

ML beats judgement; its margin over strong statistics is modest (+4.6 pts) and concentrated in history-poor SKUs

RQ2

Complements — safety-stock policy, transfers, RPA — deliver >50% of net value; data quality is the binding constraint

RQ3

Advantage holds under every single modelled shock via re-planning speed; compound shocks erase it

Treat AI as augmentation of human judgement, anchored in a well-governed ERP backbone — with privacy, ISO 27001 security and model governance as design requirements.

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Thank you — questions & discussion

What is the right counterfactual for AI-value claims?

Accuracy vs value as the evaluation construct

Simulation triangulation & composite cases as evidence

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AnyLogic model (AutoLube_Misr_AI_Forecasting.alp) and CSV output available as supplementary files